

Multiple Choice (10 marks)

1. Use partial fraction expansion to find the inverse z-transform of $F(z) = [(30z^2 - 12z) / (6z^2 - 5z + 1)]$.
 - (a) $f(n) = [3(1/2)^n + 1(1/3)^n + 1] u[n]$
 - (b) $f(n) = [4(1/2)^n - 1(1/3)^n] u[n]$
 - (c) $f(n) = [3(1/2)^n - 2(1/3)^n] u[n-1]$
 - (d) $f(n) = [3(1/2)^n + 2(1/3)^n] n[n]$
 - (e) $f(n) = [2(1/2)^n + 3(1/3)^n] u[n]$
2. The number of rings in the Bohr model of any element is determined by what?
 - (a) Number of isotopes.
 - (b) Atomic number.
 - (c) The element's period on the periodic table.
 - (d) Column number on periodic table.
 - (e) Atomic radius.
3. Find the input impedance of a 50-ohm line terminated in $+j 50$ ohms, for a line length such that $\beta l = (\pi/1)$ radian.
 - (a) 100 ohms
 - (b) 25 ohms
 - (c) $j 100$ ohms
 - (d) 50 ohms
 - (e) $-j 50$ ohms
4. A short dipole has a radiation resistance $R_r = \sqrt{(\mu_0 / \epsilon_0)} [(l)^2 / 6\pi]$ ohms. Find A_{em} the maximum effective aperture of this dipole.
 - (a) $0.075 \lambda^2$
 - (a) $0.200 \lambda^2$
 - (a) $0.180 \lambda^2$
 - (a) $0.089 \lambda^2$
 - (a) $0.102 \lambda^2$

5. Find the Fourier cosine series over the interval $0 < x < c$ for the function $f(x) = x$.

- (a) $f(x) \sim (1/2)c + (2c/\pi^2)\sum_{k=1}^{\infty} \frac{1}{k^2} \cos\left(\frac{2kx}{c}\right)$
- (b) $f(x) \sim (1/2)c + (4c/\pi^2)\sum_{k=1}^{\infty} \frac{1}{(2k+1)^2} \cos\left(\frac{(2k+1)x}{c}\right)$
- (c) $f(x) \sim (1/2)c - (4c/\pi^2)\sum_{n=1}^{\infty} \frac{1}{n^2} \cos\left(\frac{nx}{c}\right)$
- (d) $f(x) \sim (1/2)c - (4c/\pi^2)\sum_{k=1}^{\infty} \frac{1}{(2k+1)^2} \cos\left(\frac{(2k+1)x}{c}\right)$
- (e) $f(x) \sim (1/2)c - (4c/\pi^2)\sum_{k=1}^{\infty} \frac{1}{k^2} \cos\left(\frac{2kx}{c}\right)$

True / False (10 marks)

Answer True or False

- 6. True or False: The Laplace transform of $f(t) = \cos(kt)$ for $t > 0$ is $k / (s^2 + k^2)$ for $s > 0$.
- 7. True or False: The regulation of the 2,300 volt 60-cycle power transformer calculated using the per-unit method is 1.5 percent.
- 8. True or False: The constants are defined as $a' = P_c V_c^2$ and $b' = V_c / 3$ for the Dieterici equation of state.
- 9. True or False: The correct answer to 'Find the input impedance of a 50 ohm line terminated in $+j 50$ ohms, for a line length such that $\beta l = \pi$ radian.' is '75 ohms'.
- 10. True or False: A capacitance of 7,000 pF is required to resonate a coil with an inductance of 0.001 mH at a frequency of 2 MHz.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

- 11. Assuming constant field excitation, analyze how the no-load voltage of a separately excited generator changes when the speed is increased to 2000 rpm and reduced to 1600 rpm, starting from an armature voltage of 150 V at 1800 rpm.
- 12. Given a wet-bulb thermometer reading of 65° and an air temperature of 95°, analyze how to calculate the relative humidity.

End of Paper